

Lecture 6:

Phase Retrieval



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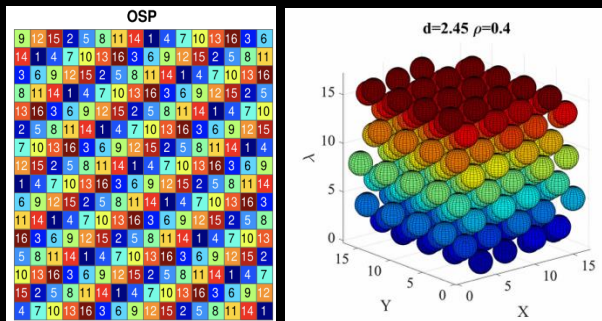
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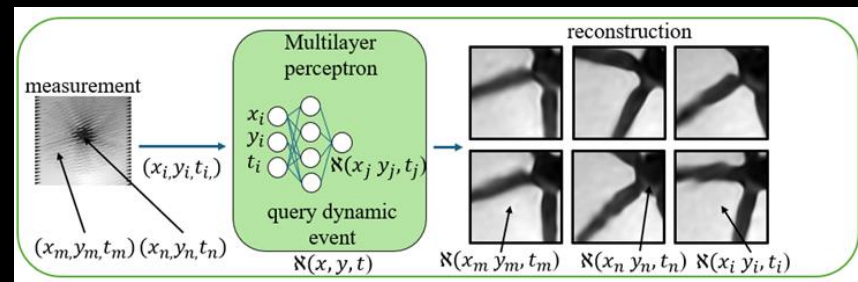
Computational Photography: Applications

- It can be applied to different signals
- It use compressive measurements
- To solve an inverse problem

Multispectral



Video

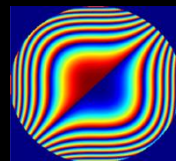


GT phase

Recovered phase

Recovered Amplitude

A woman



1995
Cubic Phase



Depth

Phase Retrieval by Designed Hadamard Complementary Coded Apertures



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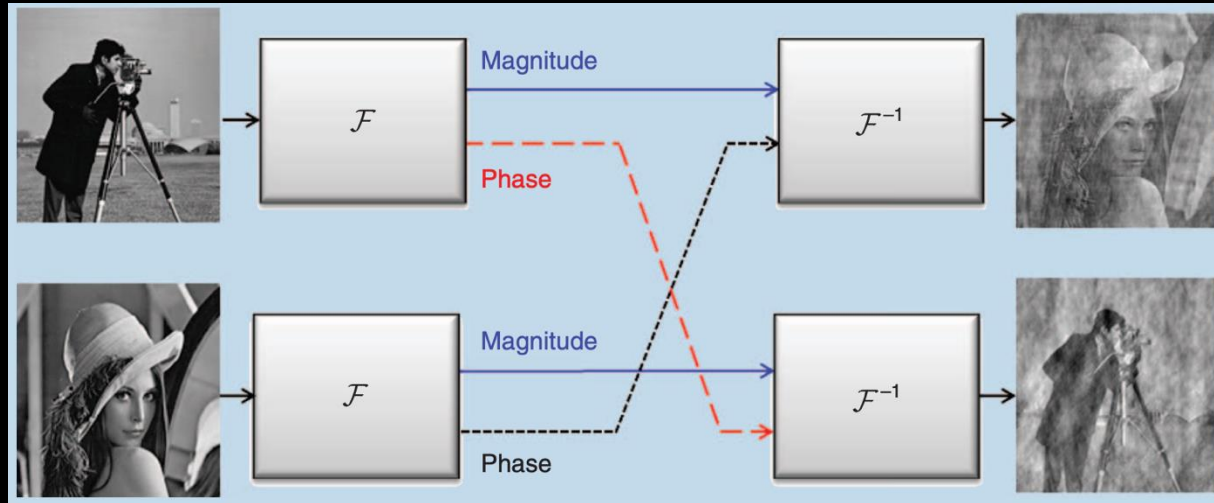
⁴ Faculty of Engineering and Applied Sciences, Universidad de los Andes, Santiago, Chile

Phase Retrieval

Phase retrieval is the problem of reconstruction of a signal or object from the magnitude of its Fourier transform, where the phase information is missing.

Why is the phase important?

The magnitude alone does not uniquely determine the signal; phase encodes crucial structural information.



The result clearly demonstrates the importance of phase information for image recovery.

Phase Retrieval

- Let $x^* \in \mathbb{C}^n$ be the unknown signal.
- Given $y_i = |\langle a_i, x^* \rangle|^2$ for $i = 1, \dots, m$
- Goal: Recover x^* (up to a global phase ambiguity)
- The problem is non-convex, making it computationally challenging
- Phase retrieval has significant applications

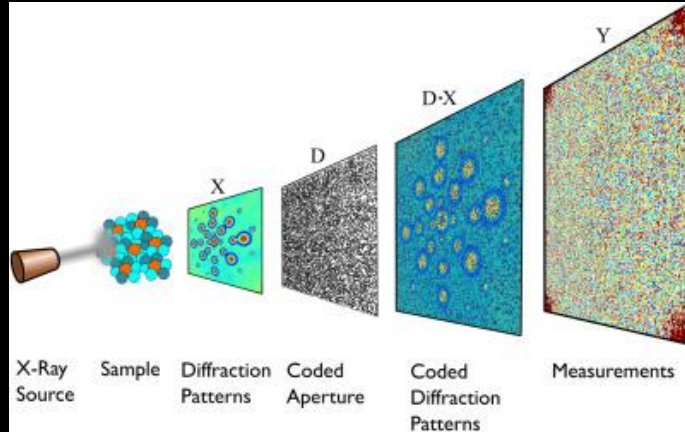
Phase Problem $y = |Ax^*|^2$

$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \left| \begin{bmatrix} a_{11} & \dots & a_{1d} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nd} \end{bmatrix} \begin{bmatrix} x_1^* \\ \vdots \\ x_d^* \end{bmatrix} \right|^2$$

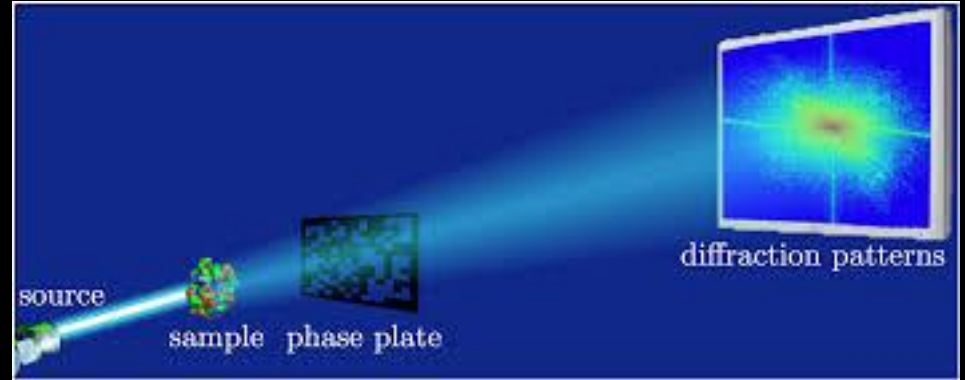
Phase Retrieval Applications

Where this happens

Many physical systems (optics, X-ray Crystallography, astronomy) can only measure intensity (i.e., squared magnitude) not phase.



X-ray-Crystallography



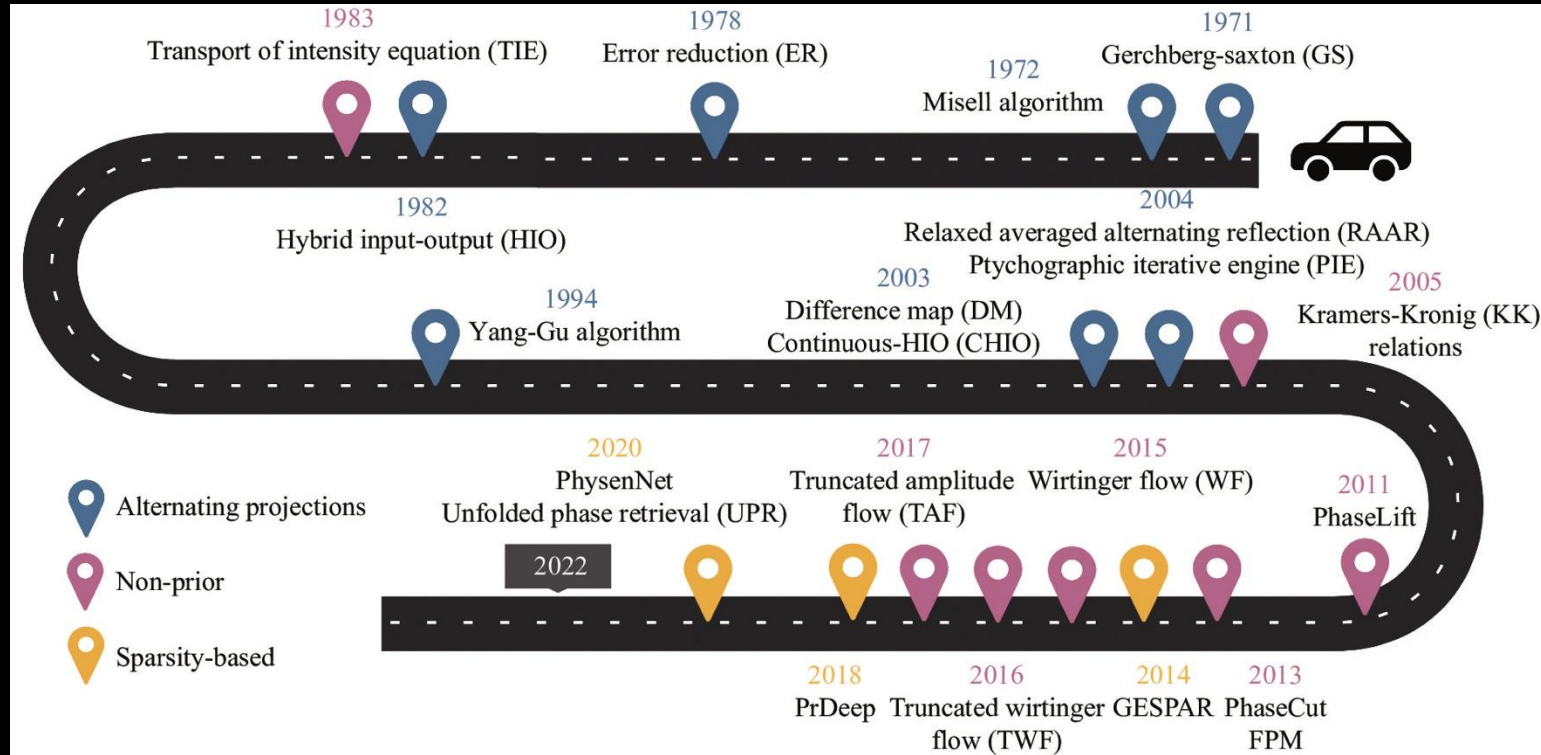
X-ray-Crystallography

Reconstruction algorithms

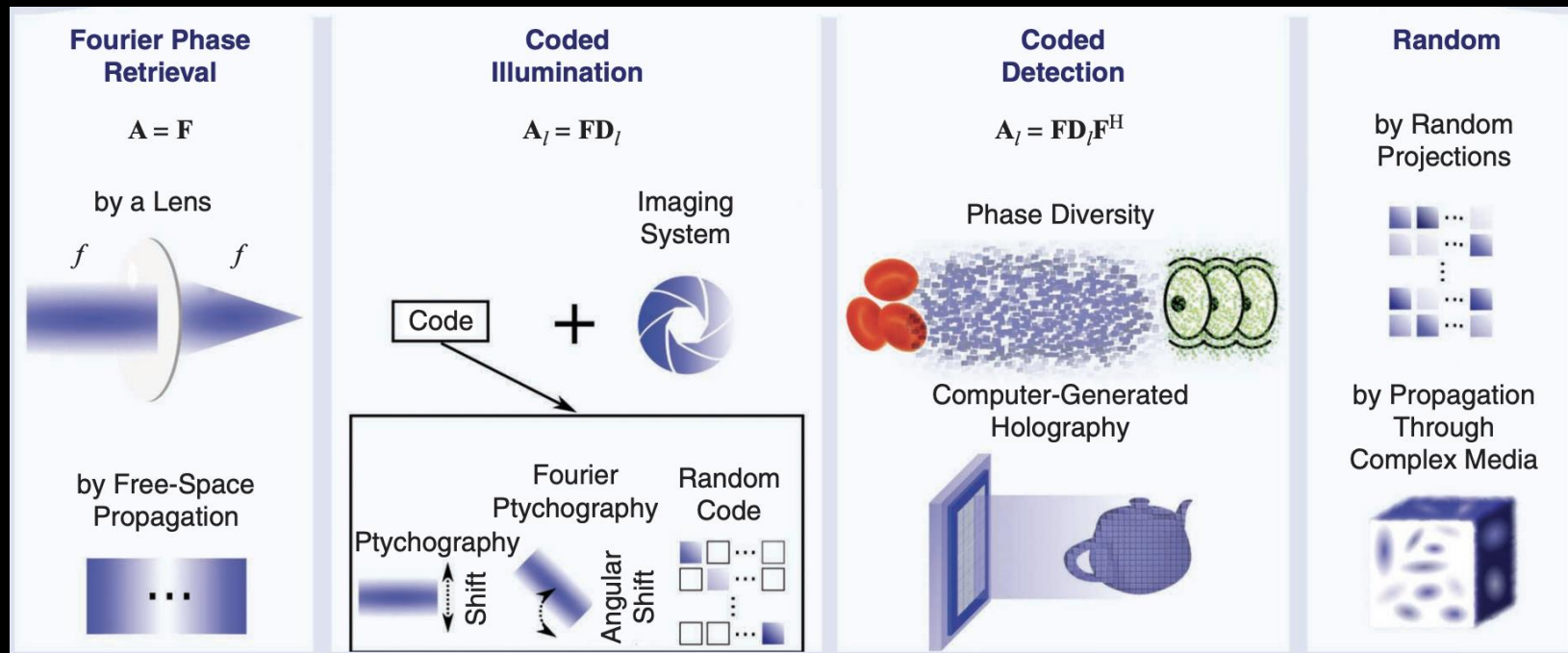
Pinilla, S., García, H., Díaz, L., Poveda, J., & Arguello, H. Coded aperture design for solving the phase retrieval problem in X-ray crystallography. *Journal of Computational and Applied Mathematics*, 338, 111-128, 2018.

Candes, E. J., Li, X., & Soltanolkotabi. Phase retrieval from coded diffraction patterns. *Applied and Computational Harmonic Analysis*, 39(2), 277-299, 2015.

Timeline Phase Retrieval Algorithms



Classes of Phase-Retrieval Problems



Coded Illumination Discrete Model

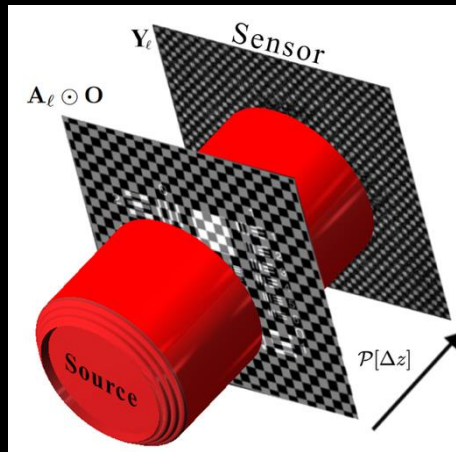
The ℓ^{th} coded measurement (CM) of the **diffraction pattern**, recorded at the image sensor is proportional to the absolute square of complex wavefront, as follows:

$$\mathbf{Y}_\ell = |\mathbf{I}_\ell|^2 + \Omega,$$

$\mathbf{I}_\ell$ is the diffraction pattern

Ω is an additive noise term

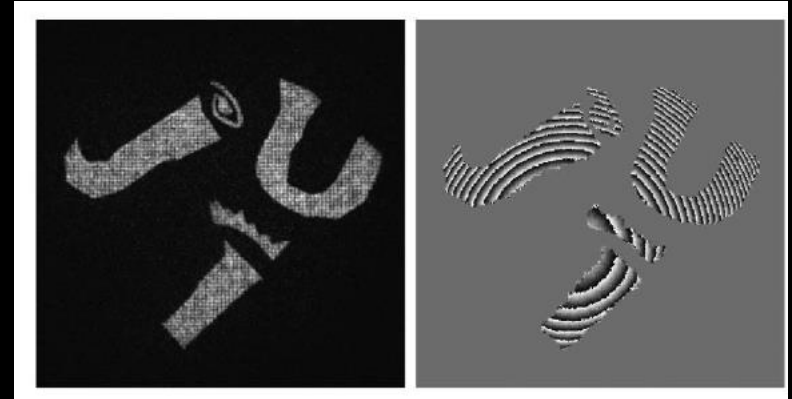
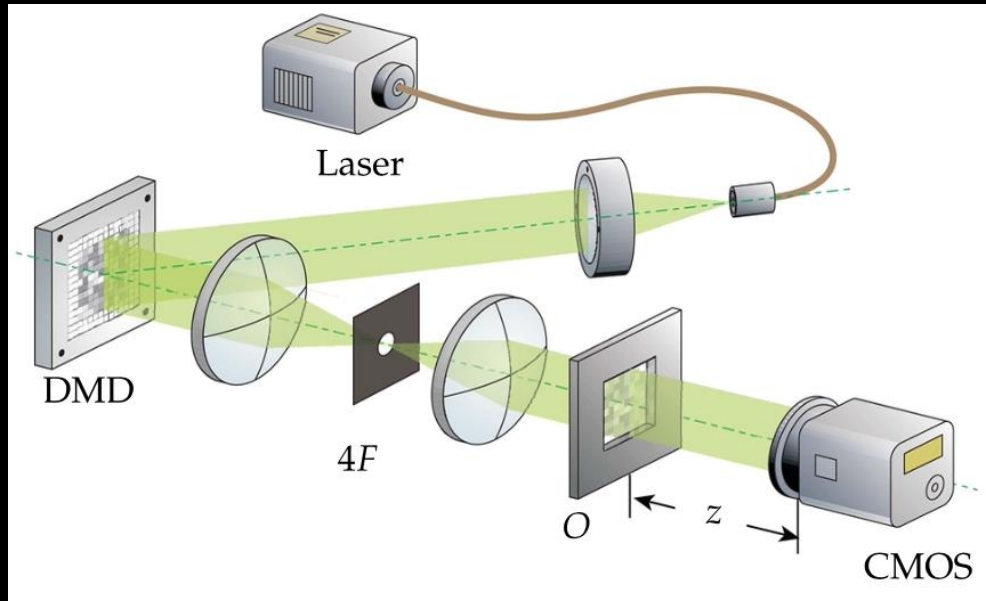
$$\mathbf{I}_\ell = \wp[\Delta z]\{\mathbf{A}_\ell \odot \mathbf{O}\},$$



- $\wp[\Delta z]$ is the operator that represents the near-field propagator.
- $\mathbf{A}_\ell$ is a set of coded apertures (CA), $\mathbf{A}_\ell \in \{\mathbf{0}, \mathbf{1}\}^{d \times d}$, with $d \times d$ denoting the number of pixels in the CA.
- A collimated coherent beam illuminates an object, producing an unknown complex wavefront $\mathbf{O} \in \mathbb{C}^{d \times d}$

Previous Work in Coded Illumination

The phase is recovered using up to 20 complementary random coded apertures, which restrict agile applications.



Amplitude

Phase

**Disadvantage: Use 20 Cas,
which also is demanding in the
reconstruction**

Peters, E., Clemente, P., Salvador-Balaguer, E., Tajahuerce, E., Andrés, P., Pérez, D. G., & Lancis, J. Real-time acquisition of complex optical fields by binary amplitude modulation. *Optics letters*, 42(10), 2030-2033, 2017.

Coded illumination in the near-field

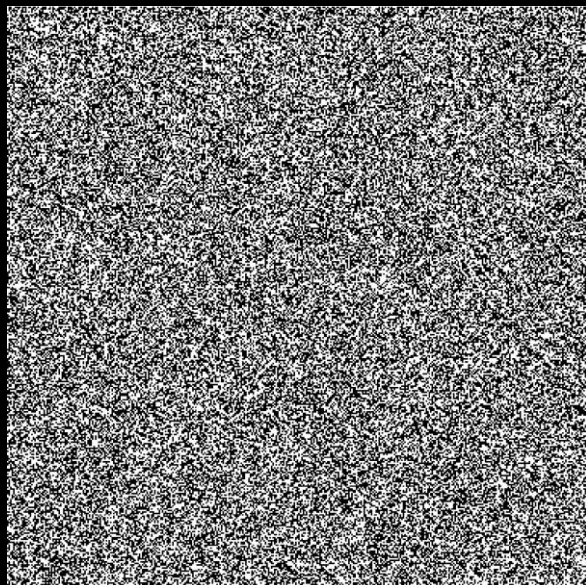
$$\mathbf{Y}_\ell = |\mathbf{I}_\ell|^2 + \Omega,$$

$$I_\ell = \wp[\Delta z]\{\mathbf{A}_\ell \odot \mathbf{0}\},$$

designing the encoding is critical
for real-time applications



measurement



Traditional random coded aperture



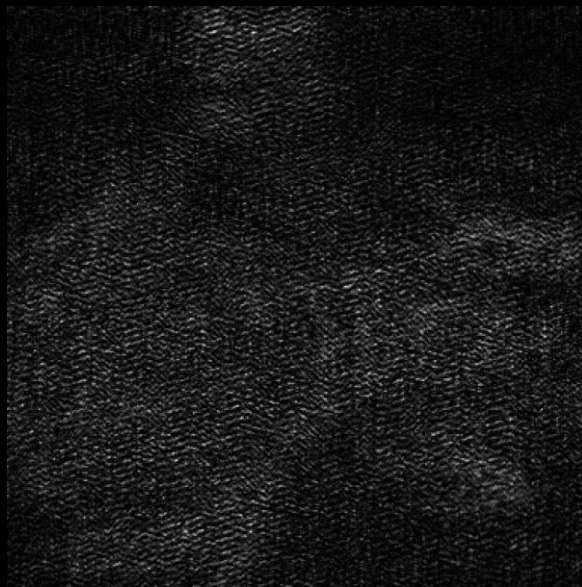
scene

Coded illumination in the near-field

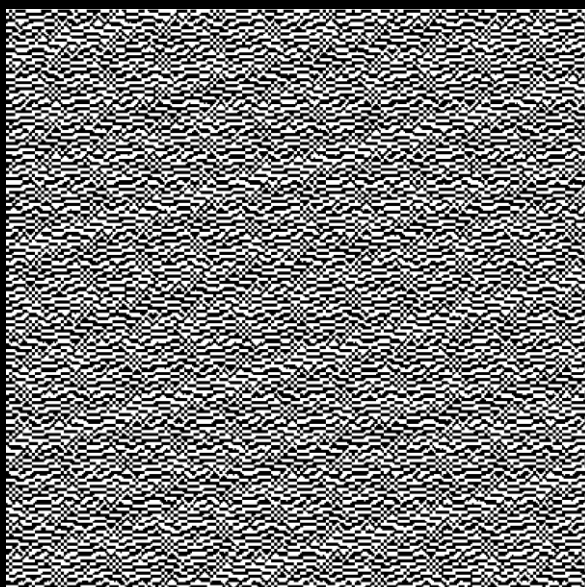
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designing the encoding is critical
for real-time applications



measurement



Designed Hadamard complementary
coded apertures



scene

Simulation

measurement

reconstructions

measurement

reconstructions



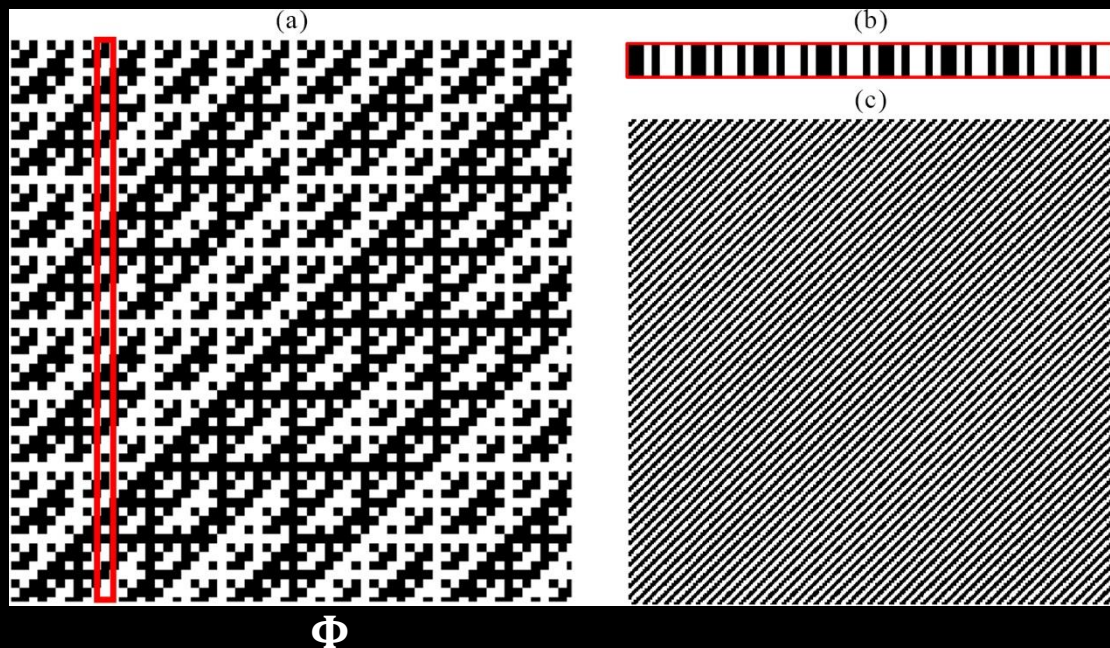
Parameters

Using 23 images of
Kodak dataset

- 16 CAs 50 iterations

Hadamard Complementary Coded Apertures (HCCA)

- The proposed coded aperture set is constructed by selecting column vectors $\phi_\ell \in \mathbb{R}^{N \times 1}$ with $\ell = \{1, \dots, L\}$ from the Hadamard matrix, optimizing for the lowest condition number. Given $\Phi_L = [\Phi_1, \dots, \Phi_L] \in \{0,1\}^{N \times L}$, we compute the covariance matrix $\Sigma = \Phi_L^T \Phi_L$ and minimize its **condition number**, $\kappa(\Sigma) = \|\Sigma\| \|\Sigma^{-1}\|$



Guarantees

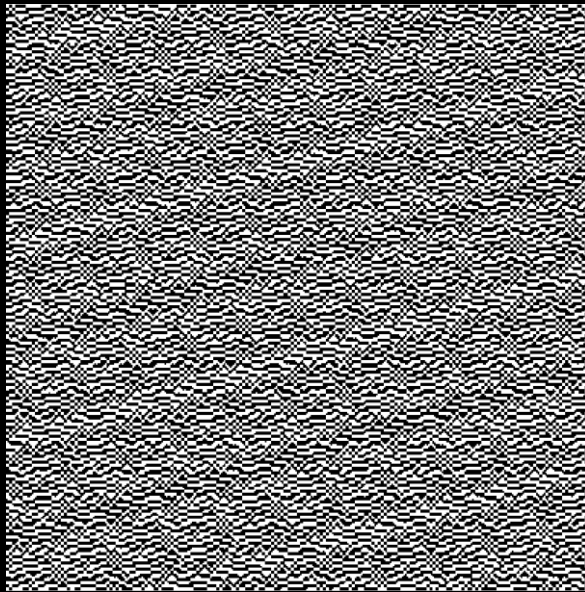
- The coded apertures exploits the a reduced condition number.
- Our design guarantees **Complementarity between pairs of coded apertures**

Hybrid Input-output Retrieval

$$\mathbf{Y}_\ell = |\mathbf{I}_\ell|^2 + \Omega, \quad I_\ell = \wp[\Delta z]\{\mathbf{A}_\ell \odot \mathbf{0}\},$$



Intensity



Designed Hadamard complementary
coded apertures



Phase

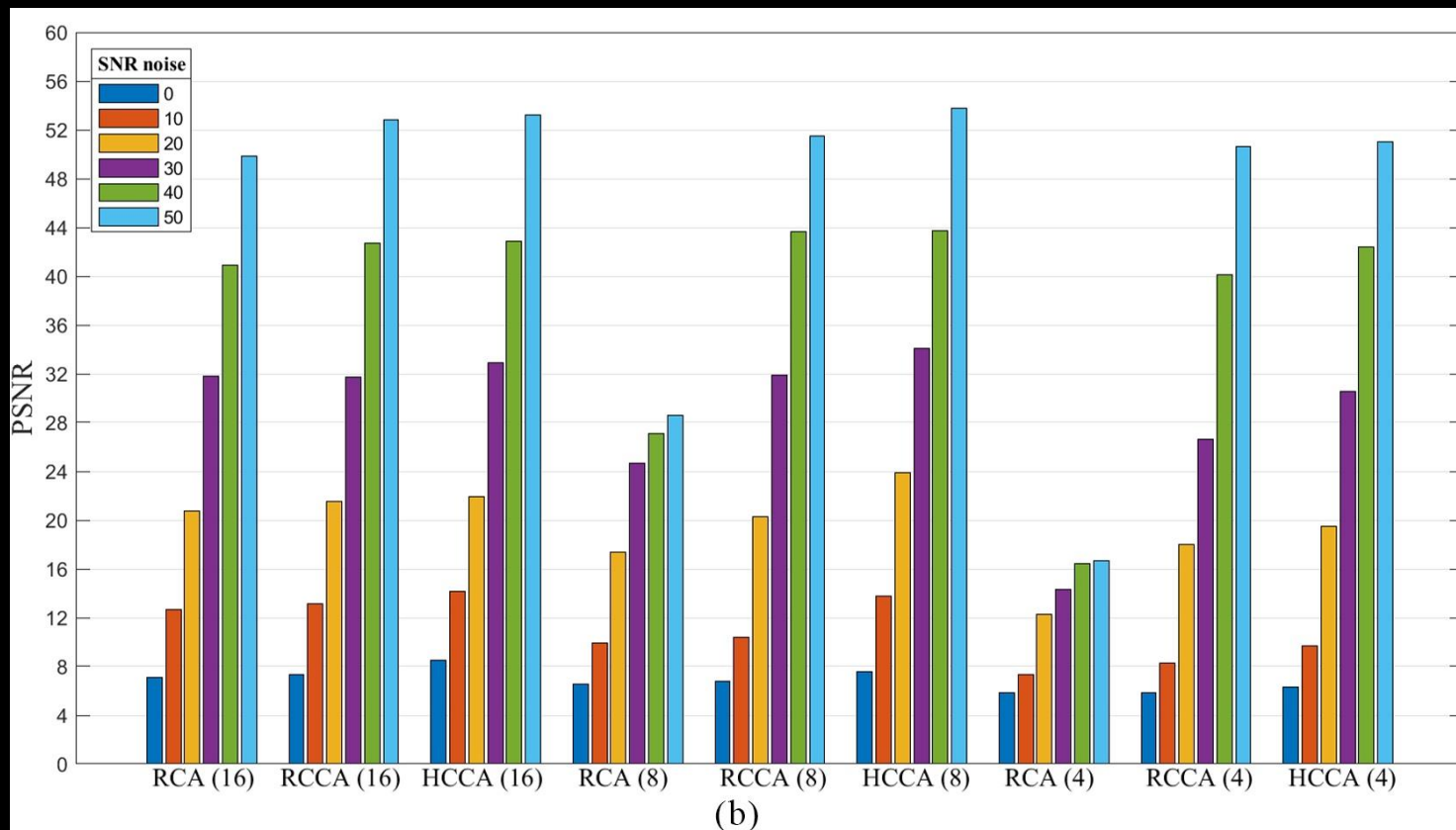
Simulation: Amplitude Quality (PSNR)

Parameters

Using 23 images of
Kodak dataset
Algorithm: HIO

- 16 CAs 50 iterations
- 8 CAs 150 iterations
- 4 CAs 300 iterations

Number of Masks	Type of coded aperture		
	RCA	RCCA	HCCA
16	9.1041	9.0965	9.0537
8	5.0229	5.0295	4.9304



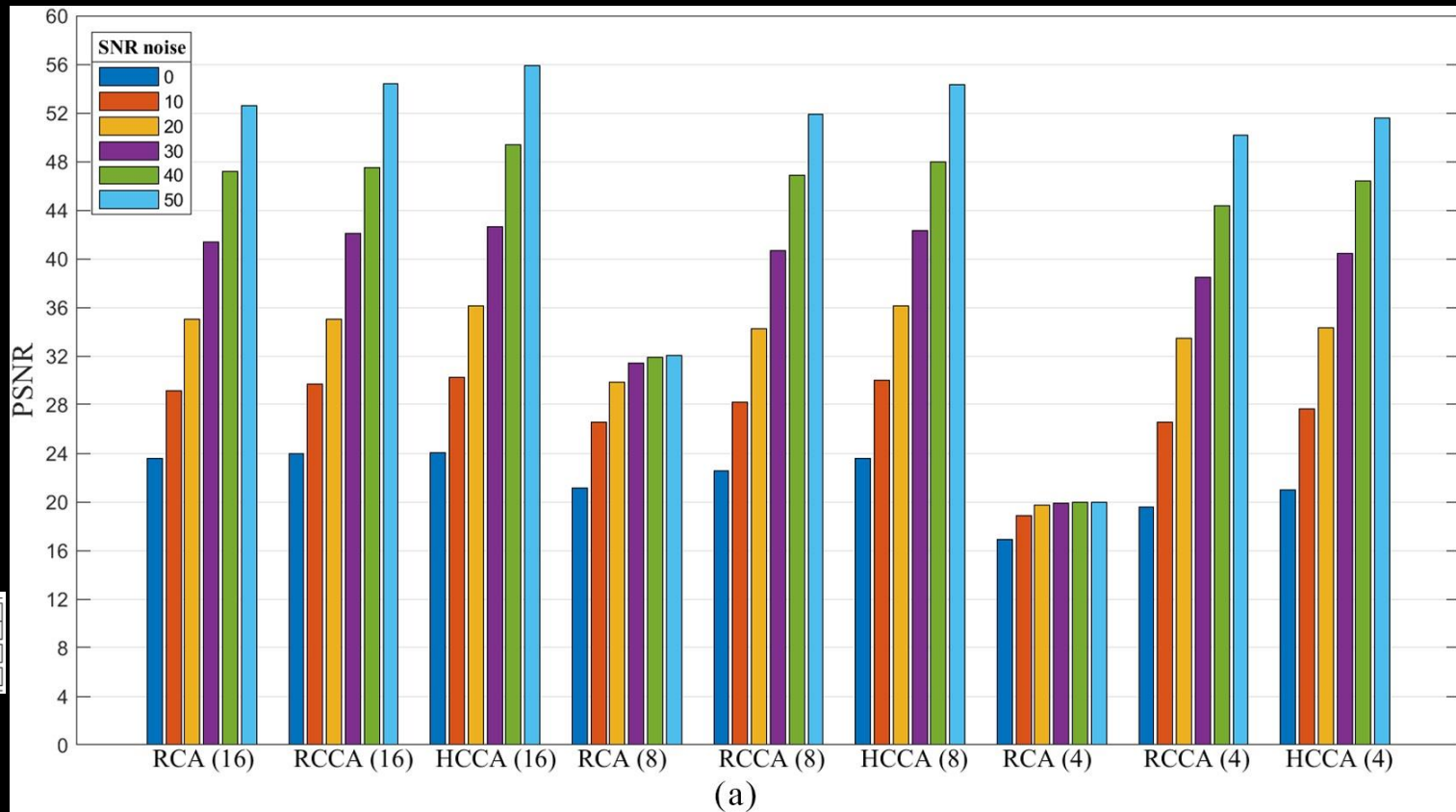
Simulation: Phase Quality (PSNR)

Parameters

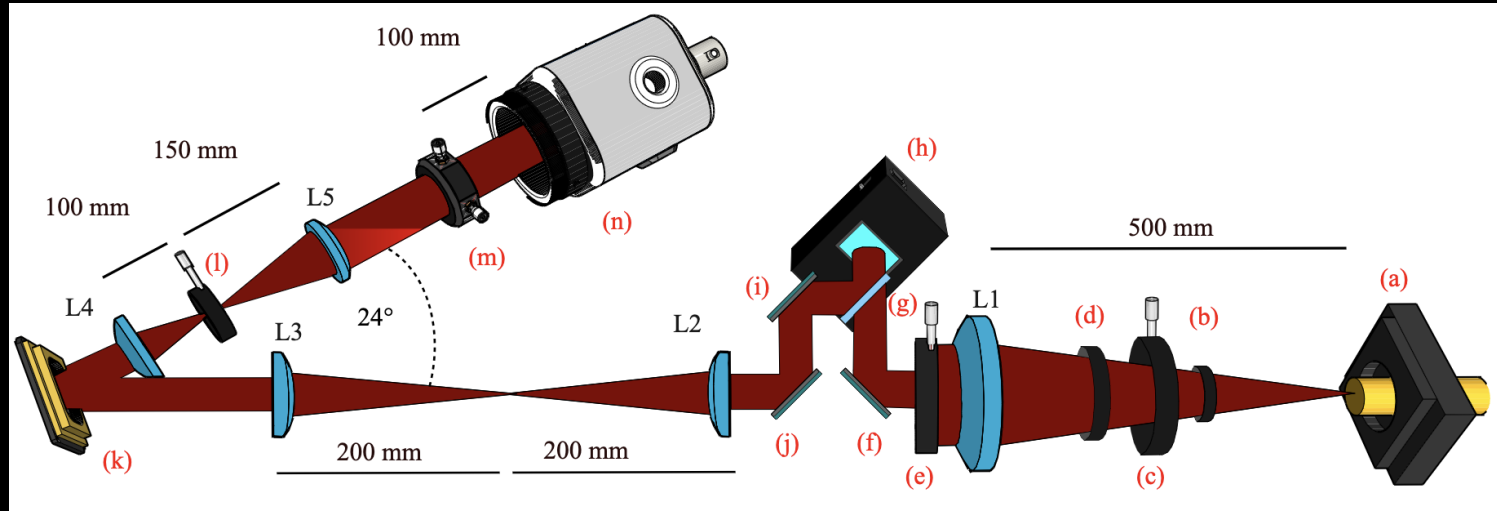
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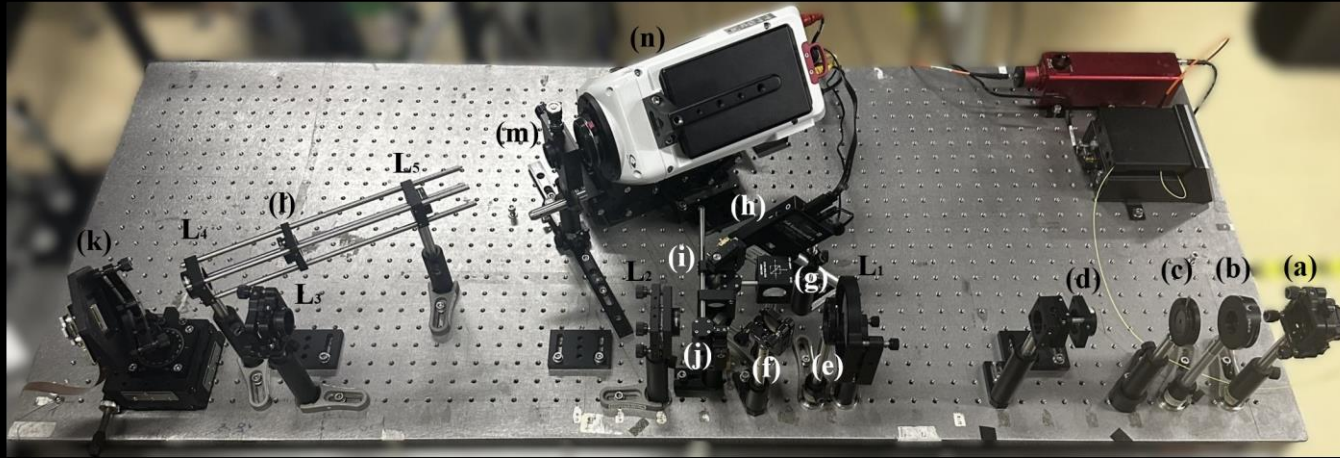


Experimental implementation



- (a) of $\lambda = 642 \text{ nm}$ propagating through different optical components as a linear polarizer (b); a half-waveplate (d); a collimator lens (L1); field stops (c), (e) and (l); mirrors (f), (i) and (j); a beam splitter (g); an SLM-LCoS (h);
- a 4f -system composed of lenses (L2) and (L3); an SLM-DMD (k); a 1.5 $\times$ telescope system composed of lenses (L4) and (L5);
- a testing object (m) placed in a plane conjugated with the LCoS and DMD devices; and finally through a CMOS camera (n)

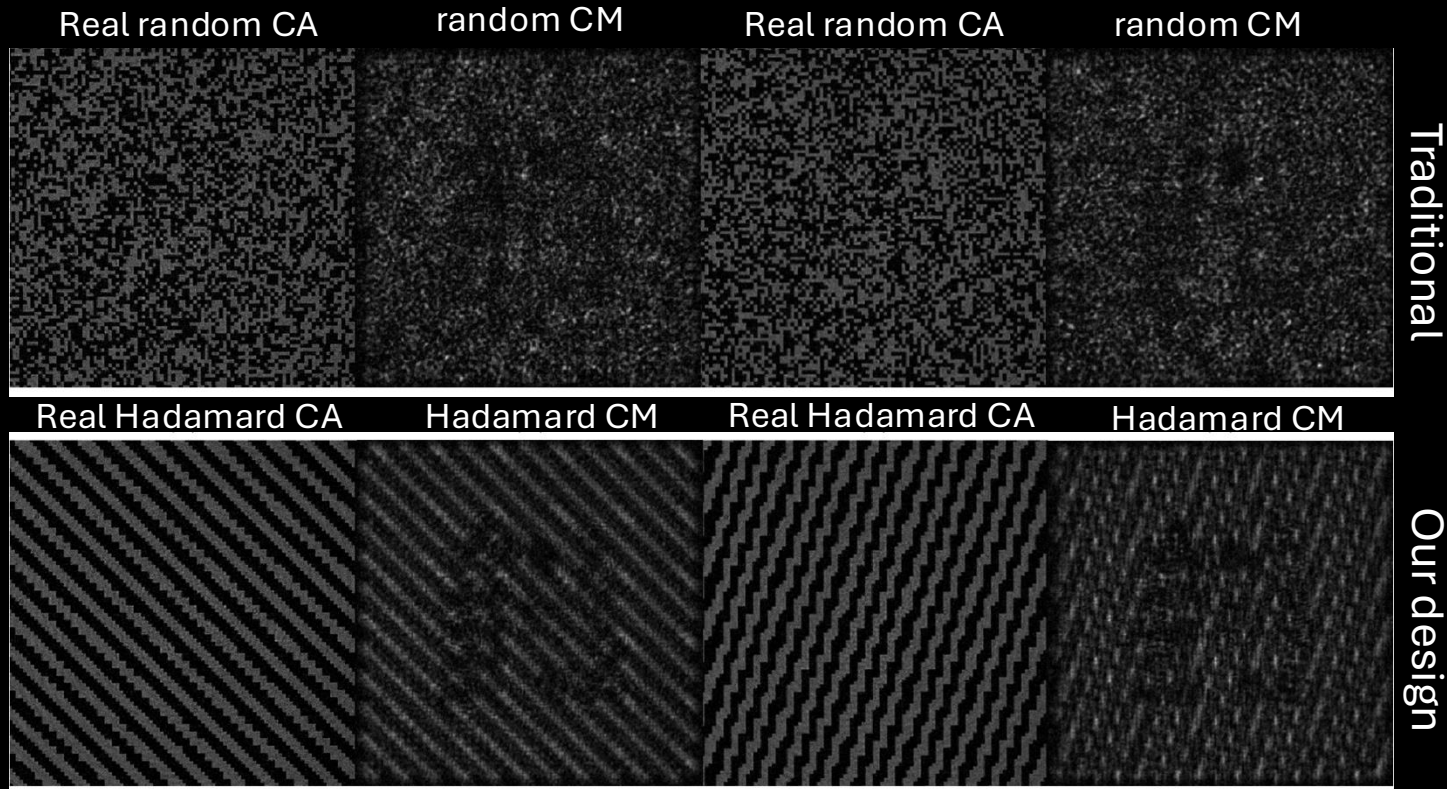
Experimental implementation



- (a) of $\lambda = 642$ nm propagating through different optical components as a linear polarizer (b); a half-waveplate (d); a collimator lens (L_1); field stops (c), (e) and (l); mirrors (f), (i) and (j); a beam splitter (g); an SLM-LCoS (h);
- a 4f -system composed of lenses (L_2) and (L_3); an SLM-DMD (k); a 1.5 $\times$ telescope system composed of lenses (L_4) and (L_5);
- a testing object (m) placed in a plane conjugated with the LCoS and DMD devices; and finally through a CMOS camera (n)

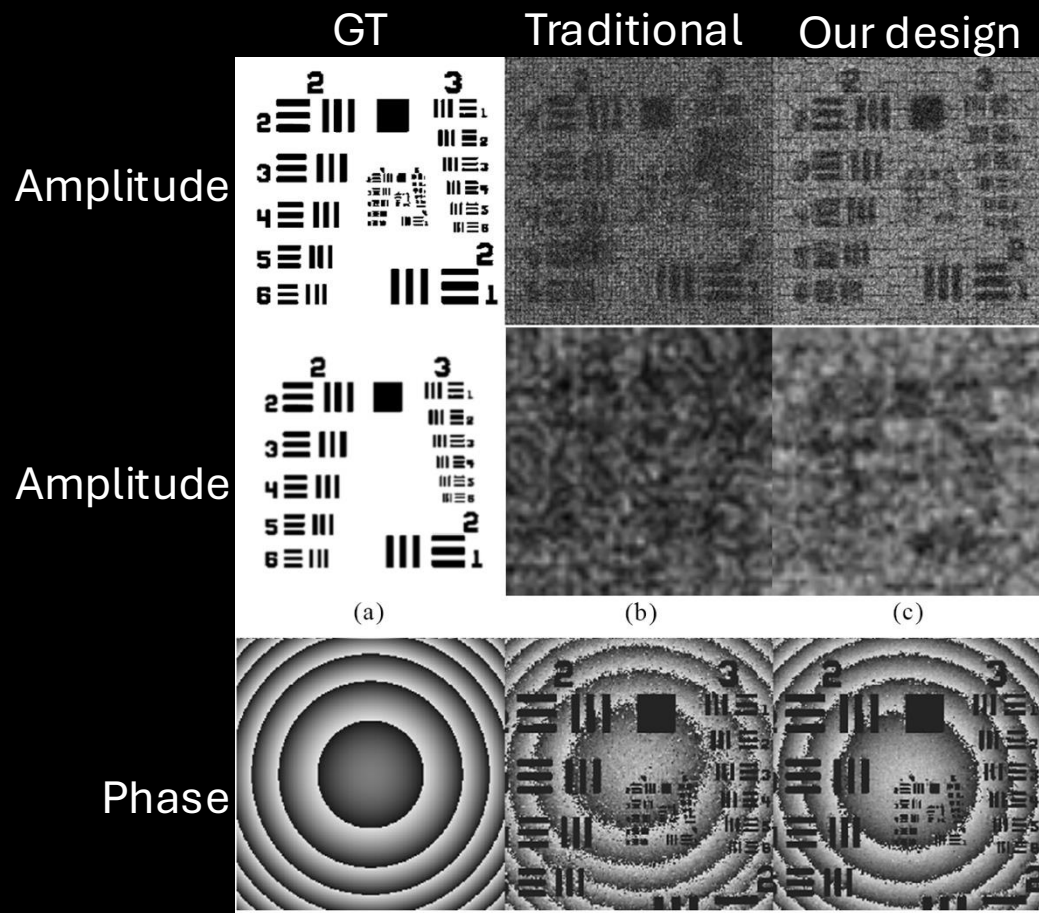
Propagation of Coded Apertures

CM: compressive measurement



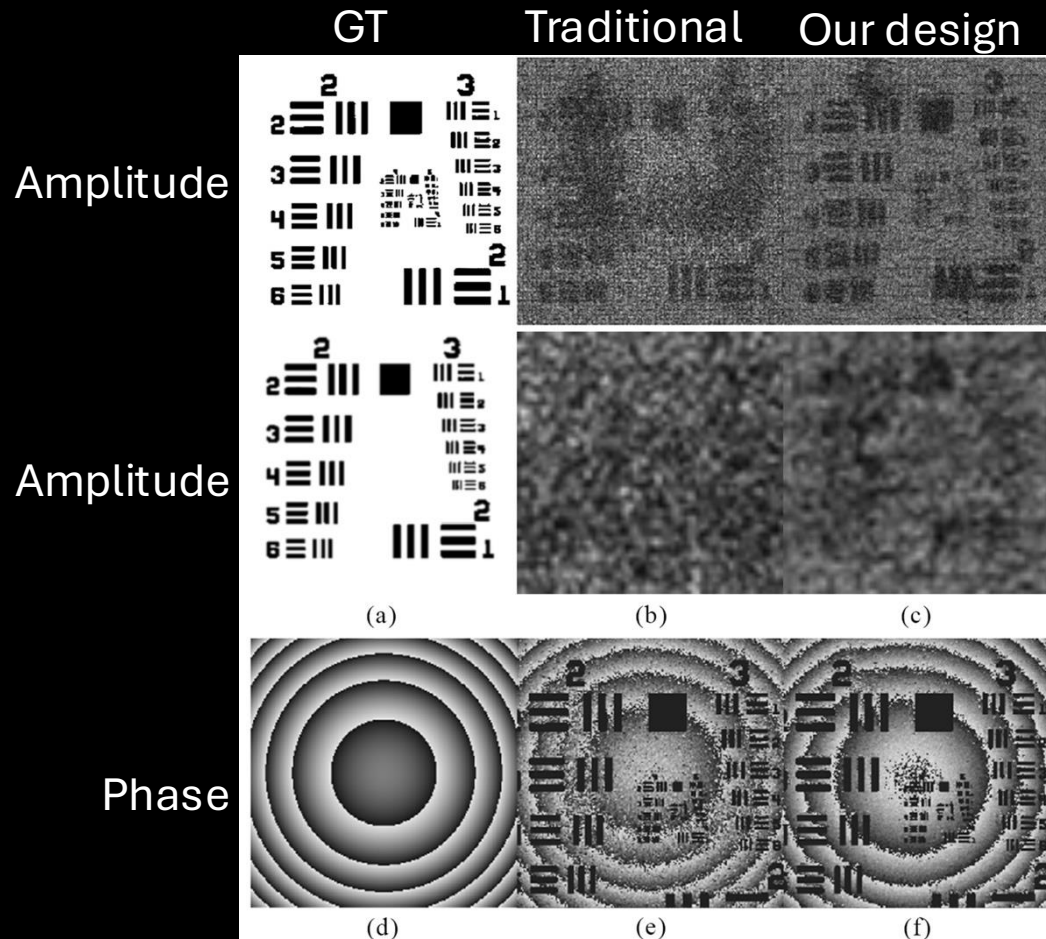
Real-data: Reconstructions

- Coded apertures: 16 RCCA, 16 HCCA
- Iterations: 100
- Reconstruction algorithm: HIO
- Amplitude: USAF 1951 benchmark
- Phase: 1000 mm lens

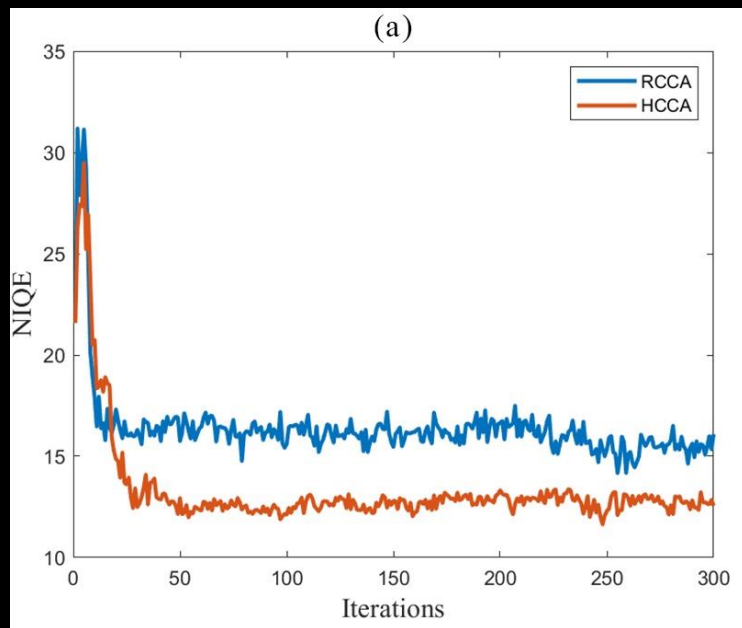


Real-data: Reconstructions

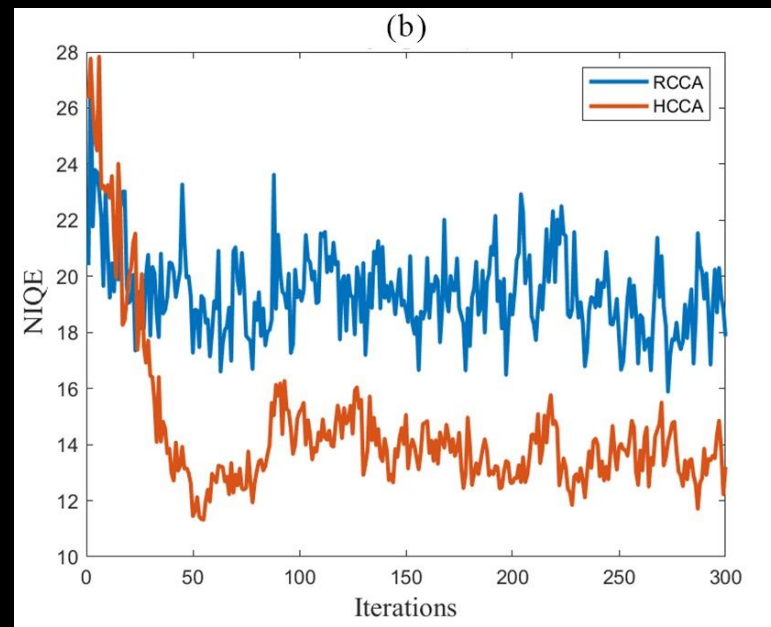
- Coded apertures: 8 RCCA, 8 HCCA
- Iterations: 1500
- Algorithm: HIO
- Reconstruction algorithm: Hybrid input-output
- Amplitude: USAF 1951 benchmark
- Phase: 1000 mm lens



Real-data: Phase and Amplitude Reconstruction



(a) Amplitude reconstruction with 16 CAs;

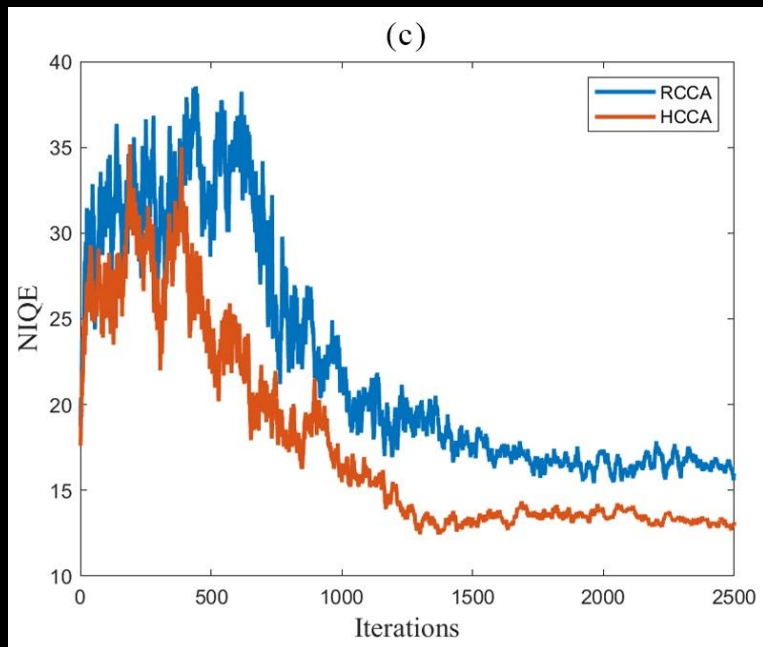


(b) phase reconstruction with 16 CAs;

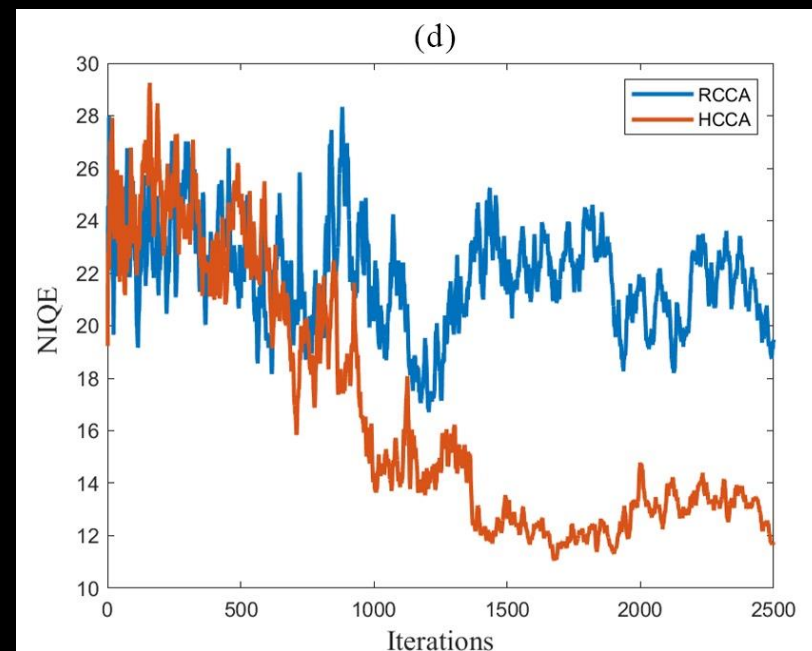
Quality comparison via the NIQE between RCCA and HCCA for 16 apertures.

NIQE: naturalness image quality evaluator metric

Real-data: Phase and Amplitude Reconstruction



(c) amplitude reconstruction with 8 CAs;



(d) phase reconstruction with 8 CAs.

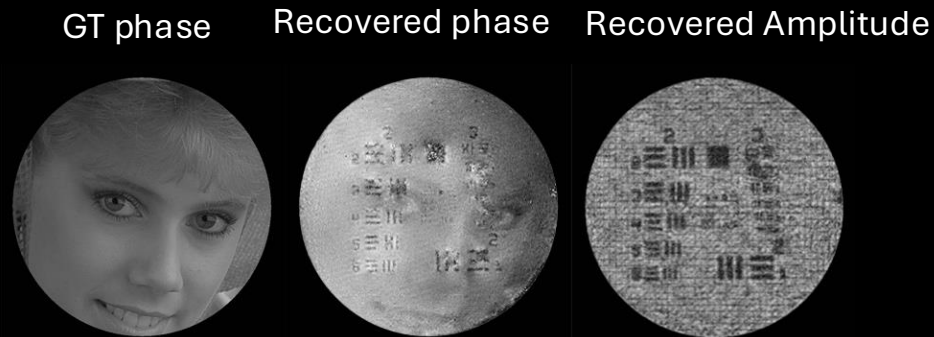
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NIQE: naturalness image quality evaluator metric

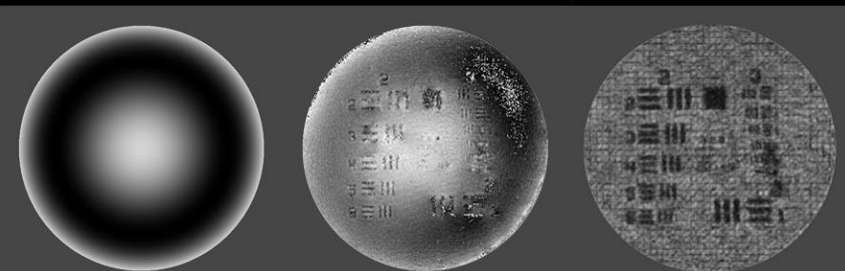
Real-data: Phase retrieval

- Coded apertures: **HCCA**
- Algorithm: HIO
- Amplitude: USAF benchmark
- Phase: Kodak image, and two Zernikes

A woman



A Zernike
Primary
spherical



A Zernike
Oblique
trefoil

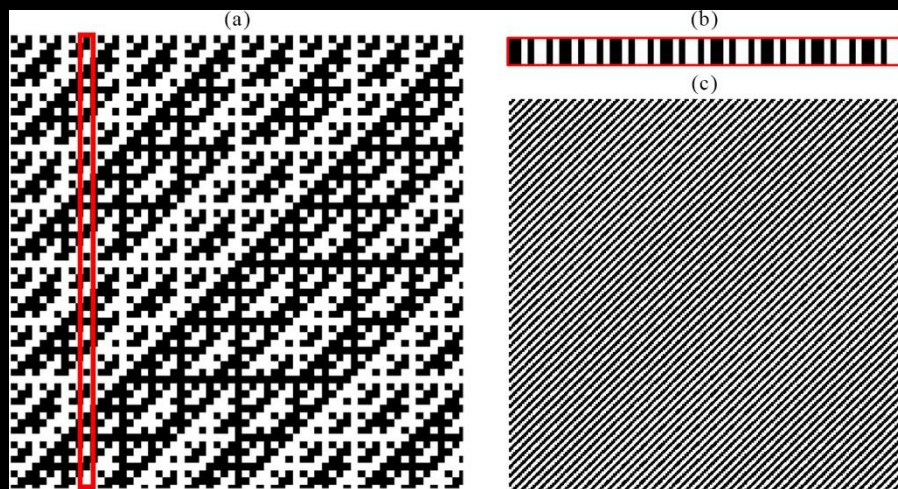


Conclusions

- We introduced a novel **Hadamard complementary coded apertures** for recovering the phase and amplitude of an unknown transmission target by encoding coherent light passing through it and measuring the intensity.

Three Advantages:

- A reduction in the number of coding masks
- Robustness to noise
- The ability to recovery arbitrary phases.
- Future work: Further reduction of mask and reconstruction algorithm



Hadamard complementary coded apertures

